Rendra Djunaedi

Robotics Software Engineer

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PROFILE

Junior robotics engineer with a Master's degree in Computer Science and Robotics, focused on autonomous systems: robot control, ROS2, perception and sensor integration. Hands-on experience controlling a UR10 arm with ROS2 (kinematics, closed-loop Cartesian control, validated on the physical robot), autonomous drone positioning, and embedded sensors (ESP32, IMU) on real hardware. Three consecutive internships building complete systems end to end, from sensors and control to supervision interface, largely autonomously. Looking to build reliable robots that operate safely in real-world industrial environments.

PROFESSIONAL EXPERIENCE

Software & Robotics Engineer, Internship (R&D / Innovation Lab) | Audensiel, Boulogne-Billancourt MARCH 2026 – AUGUST 2026

- Implemented a 3D PID controller for autonomous drone positioning (target precision 0.08 m, nominal speed 1.5 m/s) for a generic inspection mission, reusable across a 12-slot warehouse grid (3 levels, 4 slots per level).
- Built a real-time computer vision pipeline (logistics tag detection, OCR, barcode reading) with live confidence scores, using a dual 1920x1080 camera setup (20° and 90° fields of view).
- Designed and developed, fully autonomously, a drone-based logistics inspection platform (Unreal Engine 4.27, AirSim), turning an initial prototype into a fully integrated end-to-end demonstrator.
- Architected a modular, asynchronous FastAPI back-end orchestrating the simulator, robot control and vision services, with a dedicated simulation mode for hardware-independent testing, plus a React supervision interface with low-latency UDP/MJPEG video streaming.
- Built an automatic annotation pipeline (segmentation masks, up to 24 image variants per slot) for training detection models, eliminating manual annotation.

Robotics & IoT Engineer, Internship | LIRMM, Montpellier OCTOBER 2025 – FEBRUARY 2026

- Integrated inertial sensors (ESP32, MPU6050) and a computer vision module (webcam, MediaPipe, OpenCV) for real-time joint angle estimation (head, torso, shoulders).
- Built a 3D digital twin (Blender) of a human bust for posture prevention, with kinematic modeling via Denavit-Hartenberg parameters synchronized in real time with sensor data via MQTT.
- Set up a real-time data pipeline (MQTT/Mosquitto on Docker, Telegraf, InfluxDB) with Grafana dashboards for posture monitoring and history.
- Developed a Streamlit supervision interface (live monitoring and session replay) with ML-based classification of at-risk postures, validated over a continuous one-hour session with no synchronization loss.

AI / NLP Software Engineer, Internship | ESCO Lifesciences, Singapore MAY 2025 – AUGUST 2025

- Developed an email classification pipeline (7 categories) using BERT (MiniLM) embeddings and a supervised classifier, selected after a quantified comparison against SVM and Random Forest.
- Built the complete application (React, FastAPI) with secure OAuth2 / Microsoft Graph API integration and in-app model retraining (active learning loop).

PROJECTS

Visual Servoing Control of the UR10 Robotic Arm (ROS2) | Manipulation Robotics, academic project 2024 – 2025

- Modeled the forward kinematics of the 6-DOF UR10 arm (modified Denavit-Hartenberg, tool offset) and built its geometric Jacobian with tool-offset correction.
- Implemented a closed-loop Cartesian controller in position and orientation (tuned proportional gains) using a damped least-squares pseudo-inverse Jacobian to remain stable near singularities.
- Generated smooth quintic (5th-order) trajectories in position and orientation (axis-angle), giving continuous joint velocity profiles that start and end at zero.
- Validated the controller in simulation and on the physical UR10 through ROS2: position and orientation errors converged in about 2 s, and control-loop timing issues seen on the real robot were diagnosed and addressed.

QT Robot, Sensory Interaction for Robotics | LIRMM, Montpellier NOV. 2024 – JAN. 2025

Developed sensory interaction features for a robotics platform; signal processing and Machine Learning for physical interaction detection.

EDUCATION

Master's Degree in Computer Science and Robotics | Université de Montpellier 2024 – 2026

Bachelor's Degree in Computer Science | Université de Montpellier 2020 – 2024

SKILLS

Robotics & Autonomous Systems : ROS2 (UR10 arm control), robot kinematics (Denavit-Hartenberg, Jacobian), trajectory generation, PID control, autonomous navigation, SLAM, sensor fusion (Kalman filter), UAV/drones, robotics simulation
Embedded & Real-Time Systems : ESP32, NVIDIA Jetson, CUDA, MQTT, inertial sensors (IMU)
Artificial Intelligence & Computer Vision : OpenCV, object detection (YOLO), OCR, MediaPipe, PyTorch, TensorFlow, Scikit-learn
Simulation & 3D Environments : Gazebo, Isaac Sim, Unreal Engine, AirSim, Blender
Software Development : Python, C/C++, Java, C#, JavaScript
Tooling & Quality : Git, Docker, CI/CD, unit testing, CMake, Grafana
Web, APIs & Databases : FastAPI, React, REST APIs, WebSockets, PostgreSQL, InfluxDB

LANGUAGES

French : C2, Native **English** : C1, Professional Working Proficiency